

Name: \_\_\_\_\_

Solutions

Date: \_\_\_\_\_



## Subject: Year 8 Mathematics

### Test 5, 2015

#### Topics – Ratios and Rates.

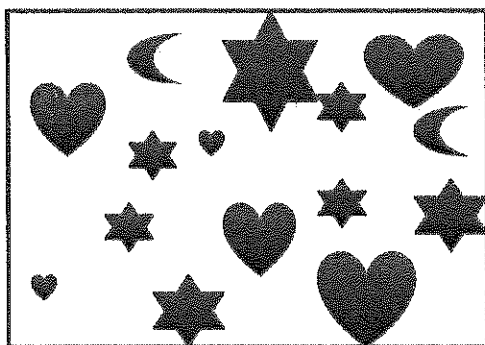
|                           |
|---------------------------|
| $\frac{\quad}{\quad}$     |
| 34                        |
| = $\frac{\quad}{\quad}$ % |

**Total Time:** 60 minutes  
**Total Reading:** 5 minutes  
**Total Working:** 55 minutes  
**Weighting:** % of the year.  
**Equipment Allowed:** ½ page notes (A4 one side)

### CALCULATOR ALLOWED

( 6 marks 3,3)

1. Complete the questions below from the pictures. (Simplify if necessary)



a)  
(i) How many Hearts

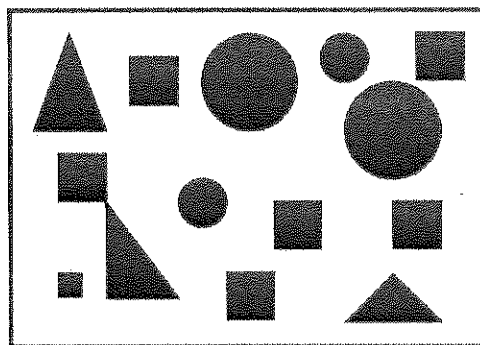
6 ✓

(ii) How many Stars

7 ✓

(iii) Hearts to Stars

6 : 7 ✓



b)  
(i) How many Squares

7 ✓

(ii) How many Shapes all together.

14 ✓

(iii) All shapes to Squares

14 : 7 ✓  
2 : 1 ✓

(- 1/2)  
if not  
Simplified

( 4 Marks, 1, 1, 2 )

2. Simplify the following ratios

a)  $4:12 = 1:3$  ✓

b)  $15:36 = 5:12$  ✓

c)  $0.3 : 0.08 = 15 : 4$  ✓✓

(1 Mark)

3. Which of the following ratios is equivalent to 20:36

a) 15 : 45

b) 15 : 18

c) 10:27

d) 15 : 27 ✓

(2 Marks, 1, 1)

4. Find the unknown values;

a)  $x : 20 = 9 : 60$

$x = 3$  ✓

b)  $6 : 4 = 36 : y$

$y = 24$  ✓

(1 Mark)

5. Which of the following is the **biggest** ratio?

a) 17 : 18 ✓

b) 8 : 9

c) 7 : 8

d) 15 : 17

( 2 Marks)

6. Tom and Bill earn some money mowing lawns. They share their earnings in the ratio 9 : 5.  
If Tom gets \$54, how much does Bill get?

a) \$84

b) \$30 ✓

c) \$14

d) \$6

(3 Marks 1, 2)

7. Orange cordial and water are mixed in a ratio of 1:4 ml

a) How much Orange Cordial do I need to add to 25ml of water?

$$6.25\text{ml} \quad \checkmark$$

b) How much water is needed to make a mixture of 150ml

$$\frac{150}{5} = 30 \times 4 = 120\text{ml} \quad \checkmark \quad \checkmark$$

(2 Marks)

8. Natalie and Chloe had \$463, which they wished to share in a ratio of 3 : 2. How much money did Natalie receive?

$$\frac{463}{5} \times 3 = \$277.80 \quad \checkmark \quad \checkmark$$

(3 Marks)

9. Concrete is made using sand, gravel and cement mix in a ratio of 2 : 3 : 1. If 78 shovels of ingredients are used altogether how much sand and gravel was used?

$$\frac{78}{6} = 13 \quad \checkmark$$

$$\text{Sand} = 13 \times 2 = \underline{26} \quad \checkmark$$

$$\text{Gravel} = 13 \times 3 = \underline{39} \quad \checkmark$$

(2 Marks)

10. Sally has gone to the hairdressers to have her hair dyed pink! The hairdresser needs to mix hair colour to developer to a ratio of 4:3. If Sally's hairdresser uses 21ml of developer, How much hair dye will she need?

$$\times 7 \left( \begin{array}{l} 4:3 \\ 28:21 \end{array} \right) \times 7 = \underline{28\text{ml}}. \quad \checkmark$$

(1 Mark)

11. A boat travels at an average speed of 50 km/h. How long will it take to travel 150 km?

$$\frac{150}{50} = 3 \quad 3 \text{ hours} \quad \checkmark$$

(2 Marks)

12. A factory produces 64 cars in a 8 – hour day. Due to a malfunction, one day the factory worked 5 hours. How many cars will be made?

a) 61

b) 40

c) 48

d) 32

$$\frac{64}{8} \times 5 = 40$$

(2 marks, 1,1)

13. From the following, which is the best buy? (circle the best option)

a)

i) 860 pens for \$90

or

ii) 270 pens for \$150

b)

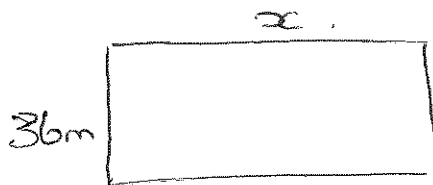
i) 352 pens for \$180

or

ii) 96 pens for \$54

(3 Marks)

14. Find the perimeter of a rectangle if its width is 36m and the ratio of length to width is 7 : 6



$$x = 42$$

$$\times 6 \quad \left( \begin{array}{l} 7 : 6 \\ x : 36 \end{array} \right) \times 6$$

$$\text{Perimeter} = (42 \times 2) + (36 \times 2)$$

$$= 156\text{m}$$